Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

Gajavada Sanjeevkumar

Problem Statement

Question

Draw a **cycloid** generated by a circle of diameter $D = 50$ mm rolling along a straight base line for **one complete revolution** without slipping.

Construct a **tangent** and a **normal** to the cycloid at any point P located on the curve.

Step 1: Draw Generating Circle & Directing Line

$C_0 \longrightarrow$ Initial Center Point
$AA' \longrightarrow$ Directing Line
Axis $\longrightarrow$ Center Line

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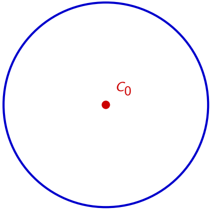

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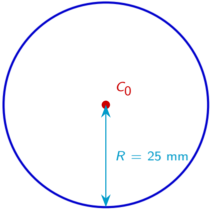

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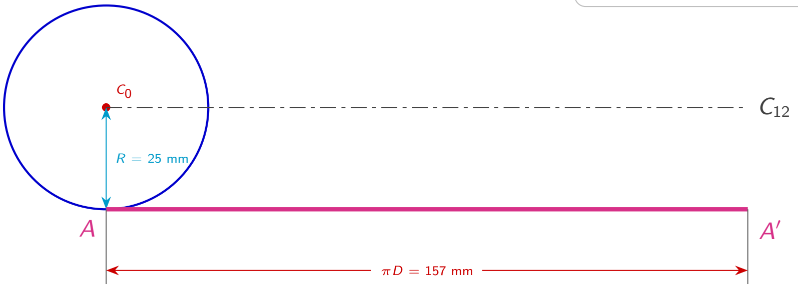

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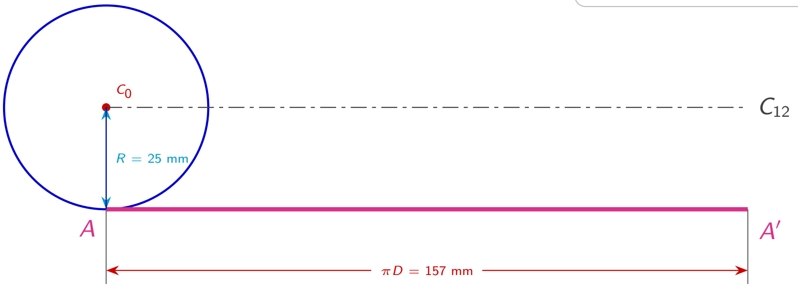

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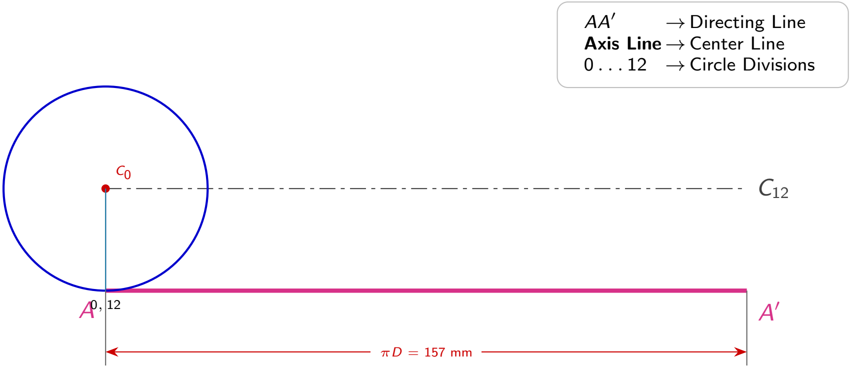
Step 2: Divide Generating Circle into 12 Equal Parts



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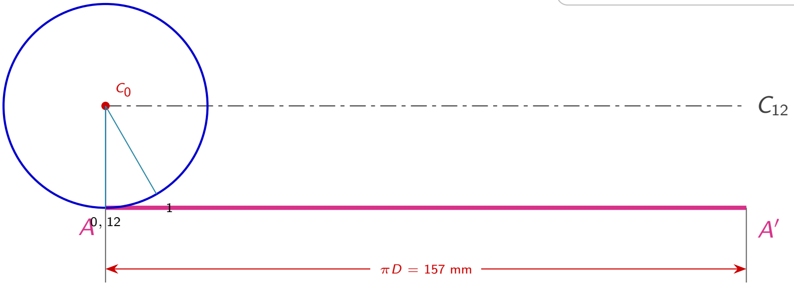

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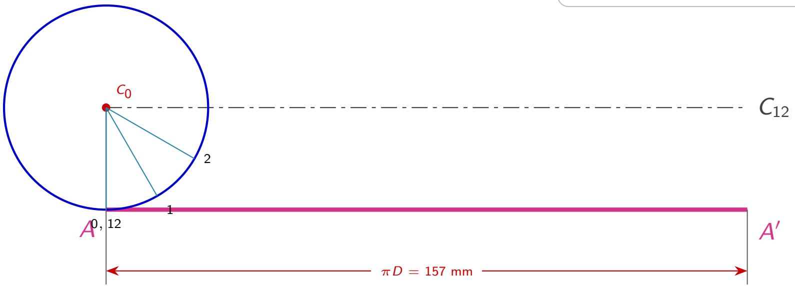

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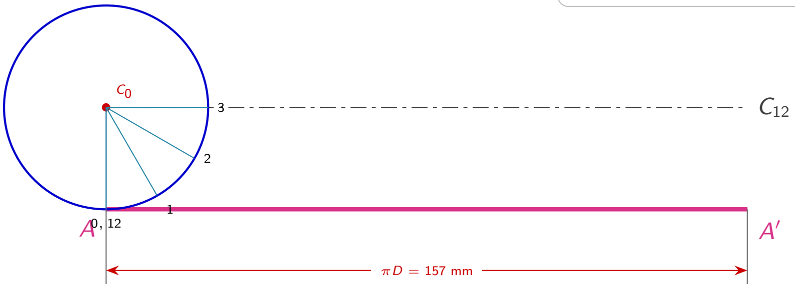

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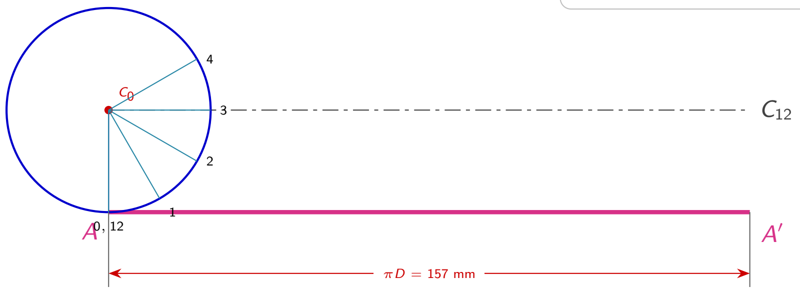

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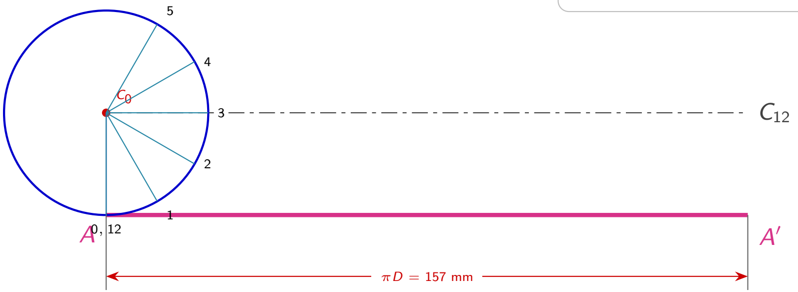

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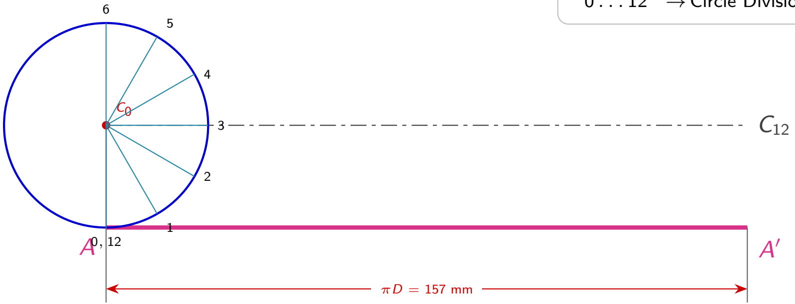

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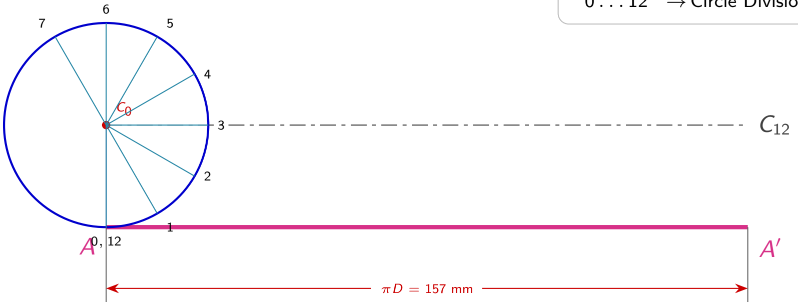

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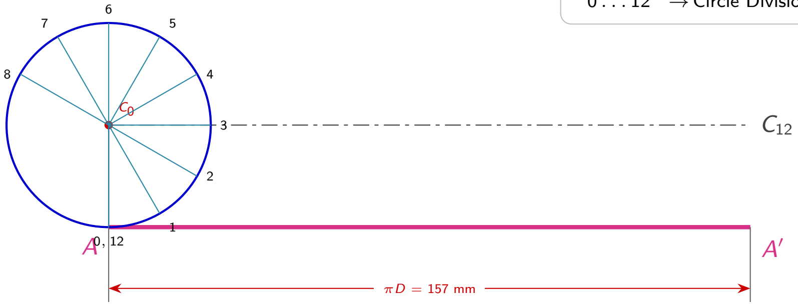

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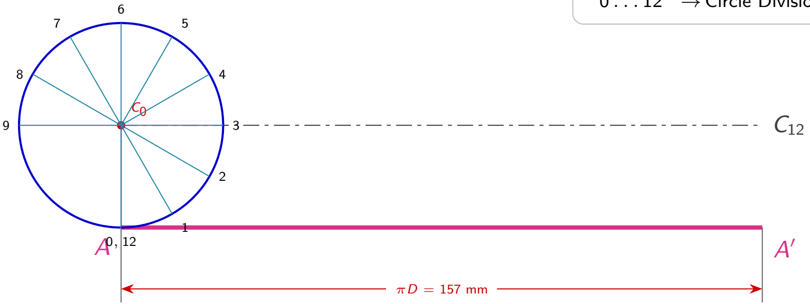

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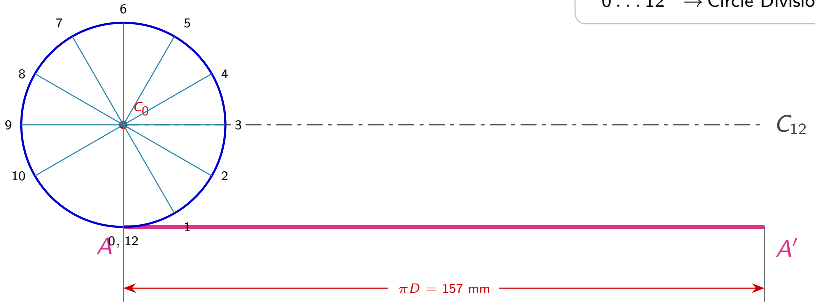

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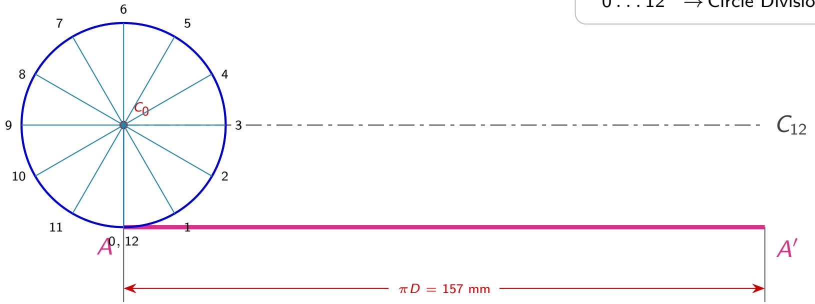

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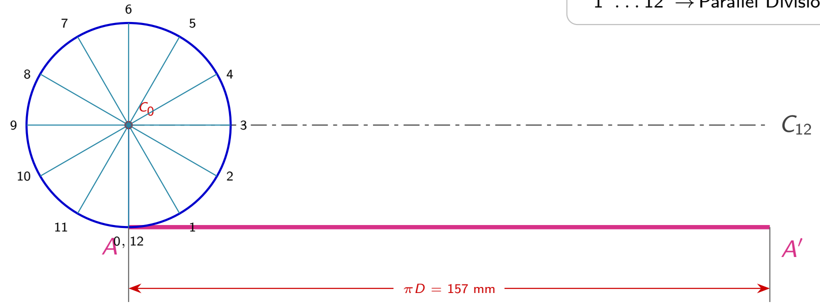
Step 3: Divide Directing Line using Line Division Method



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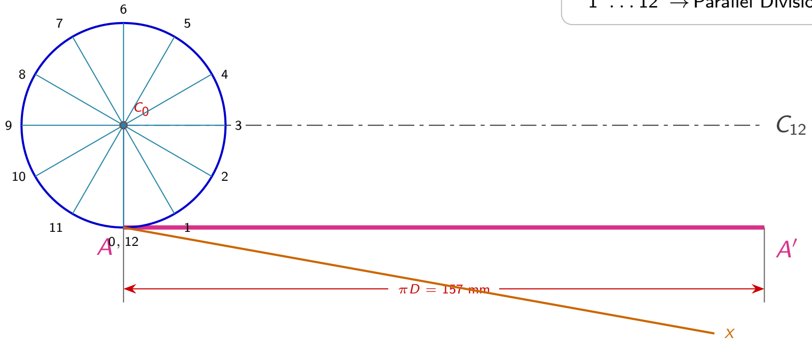

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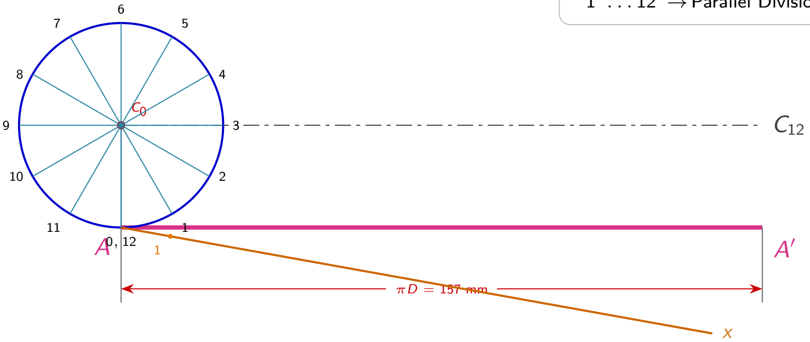

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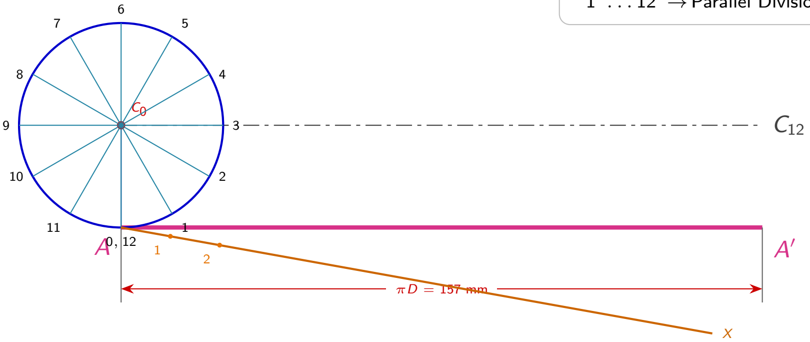

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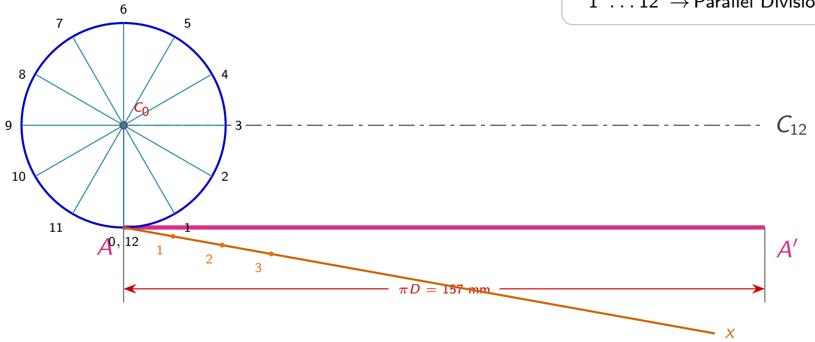

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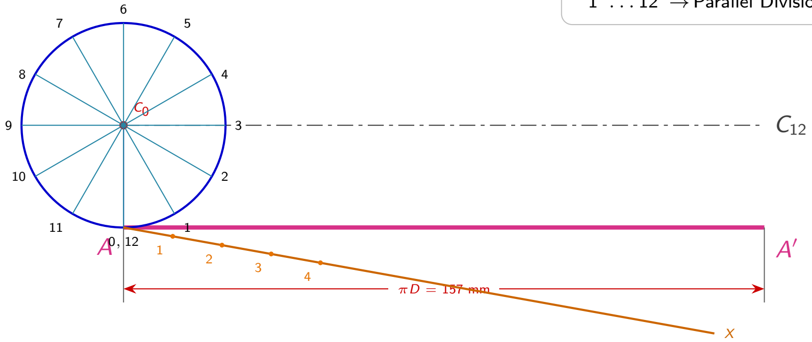

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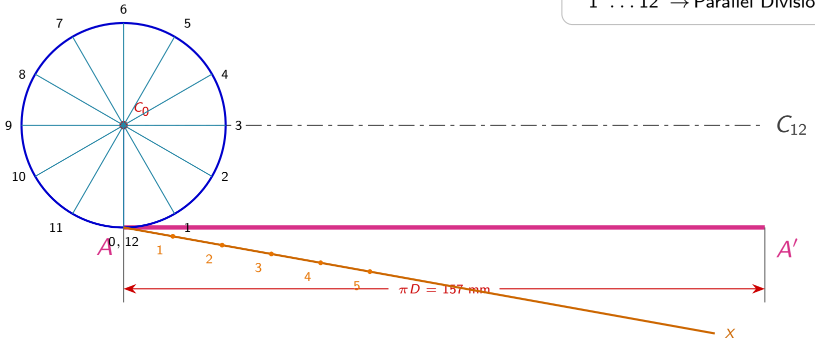

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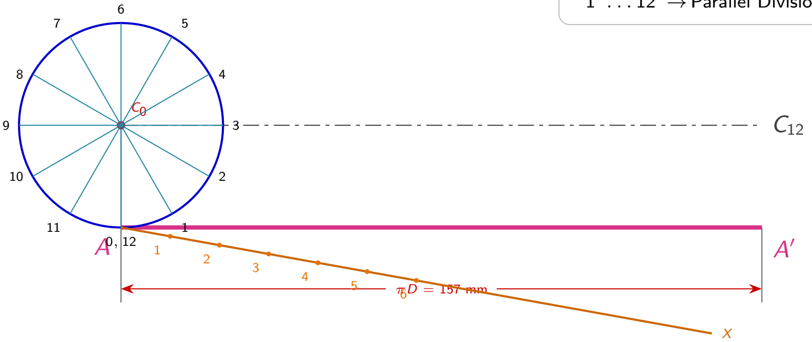

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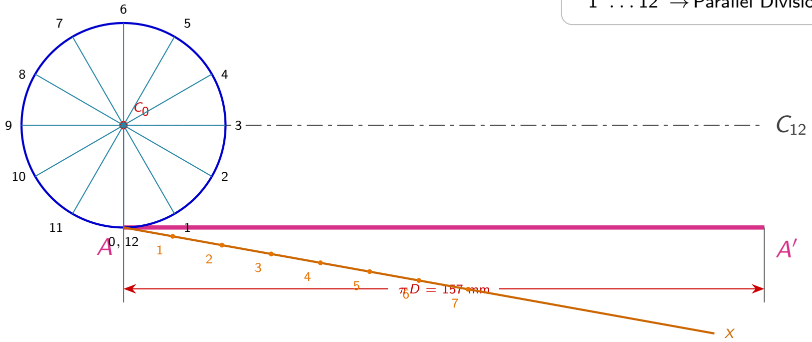

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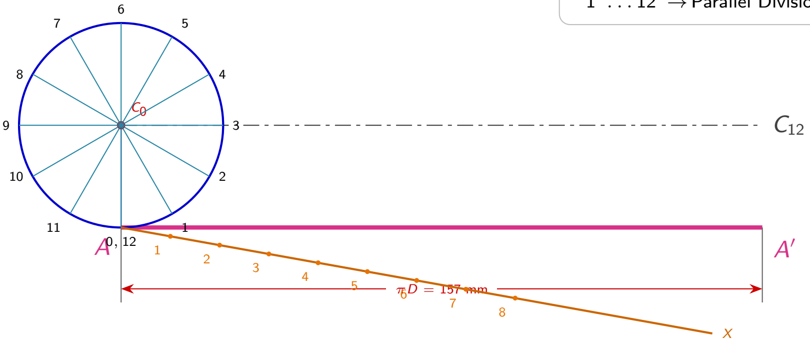

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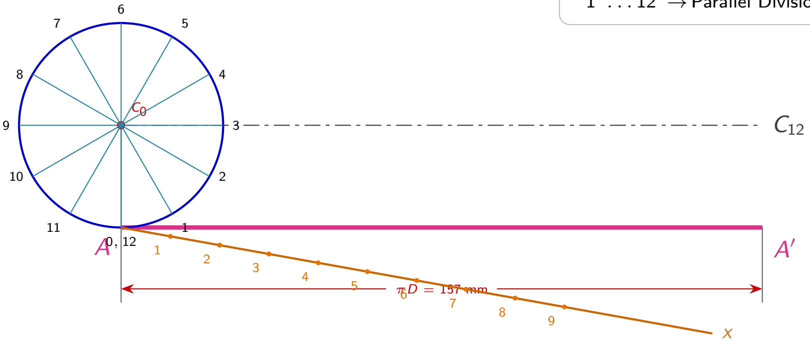

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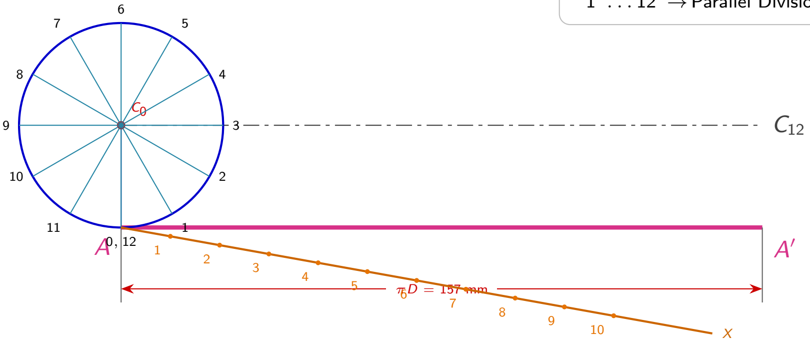

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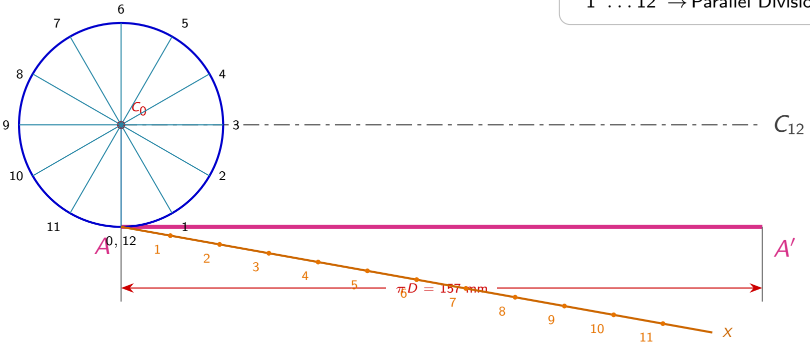

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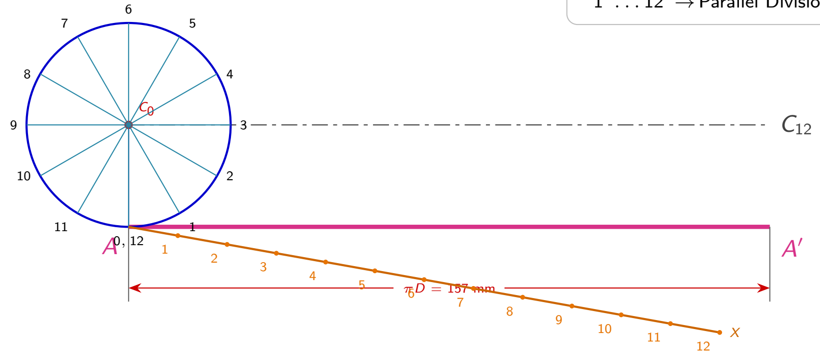

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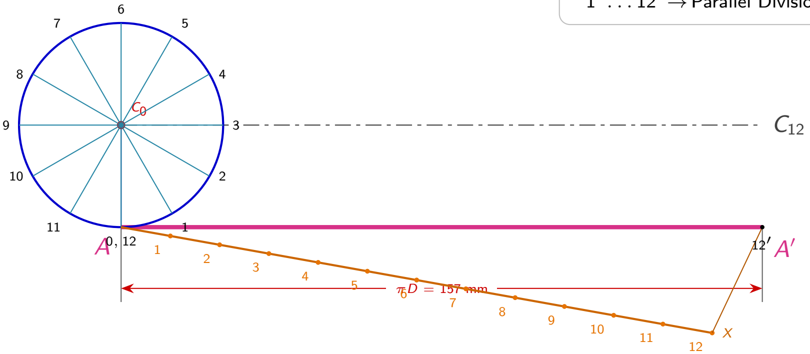

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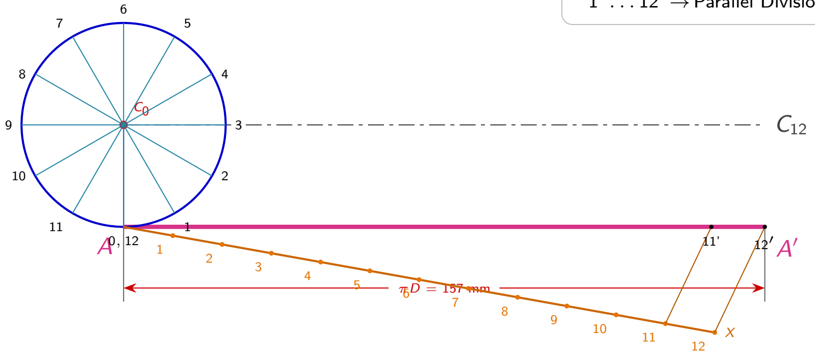

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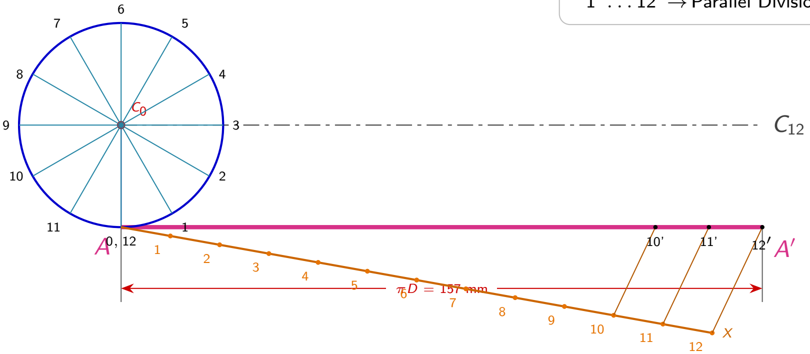

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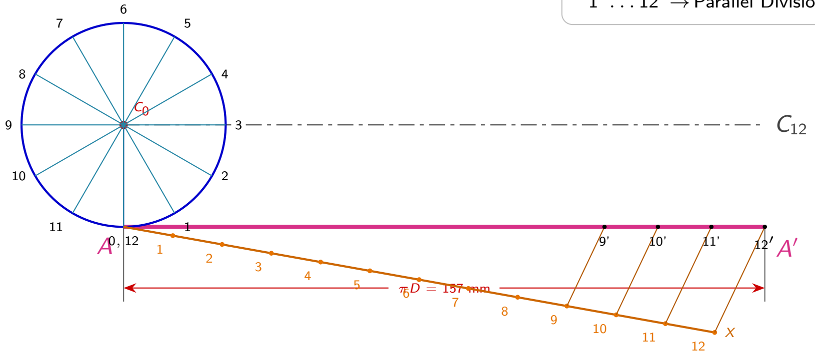

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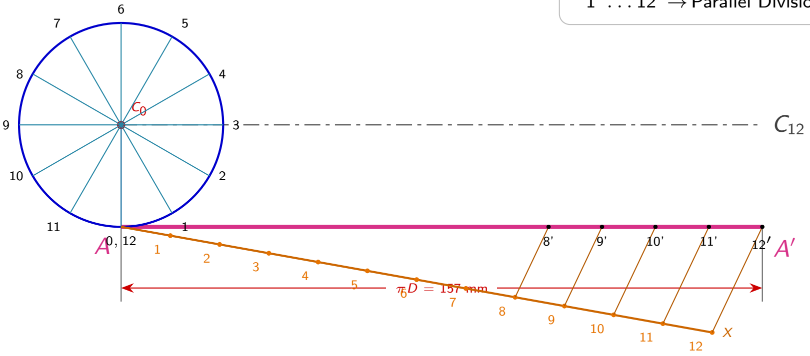

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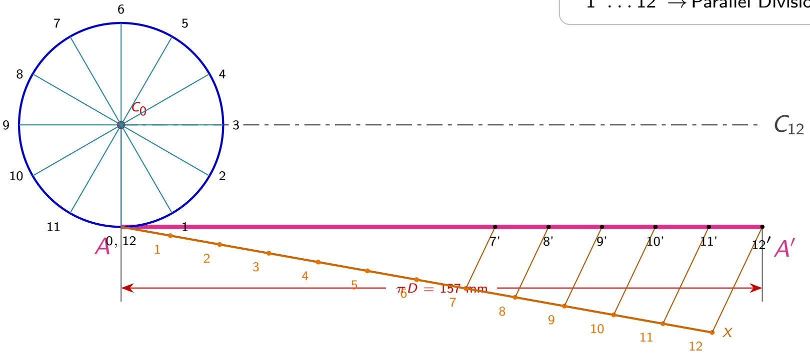

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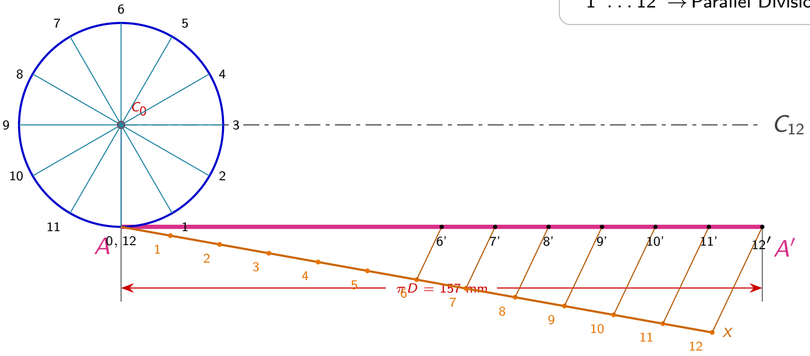

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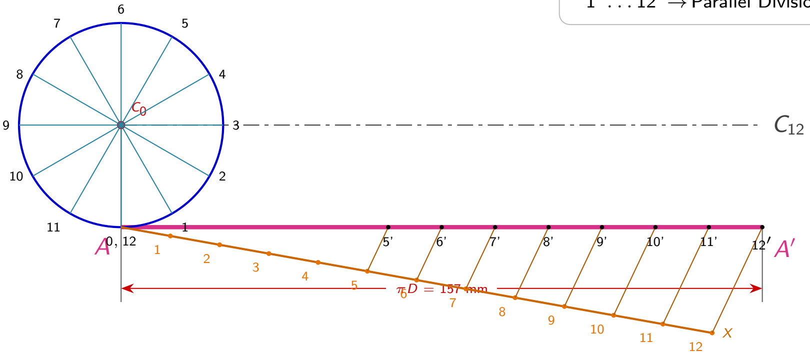

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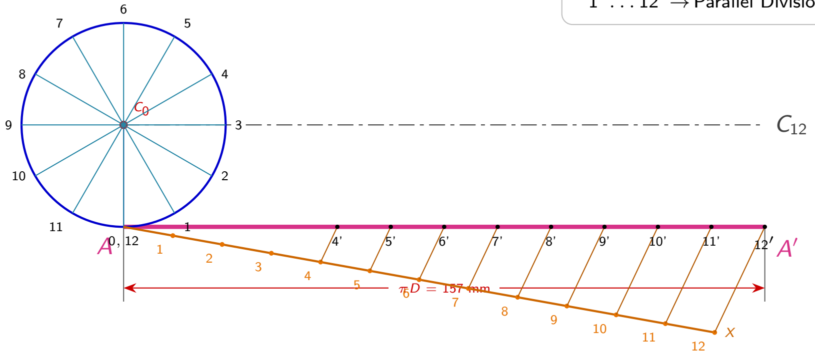

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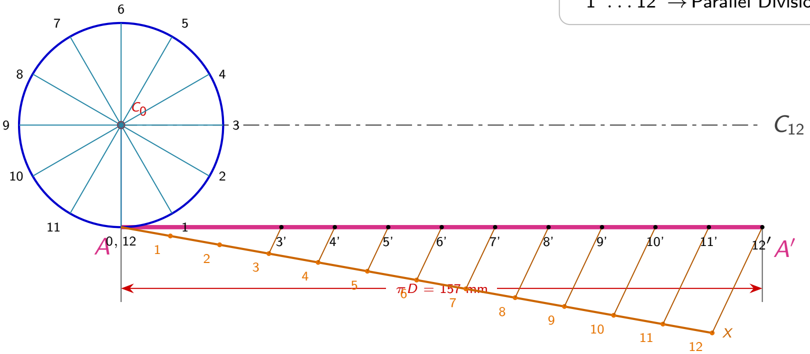

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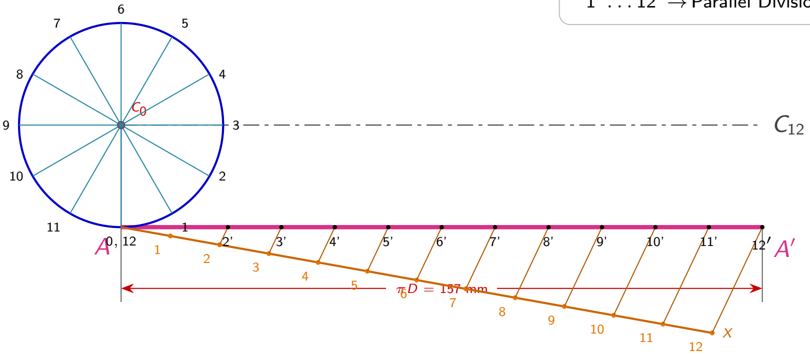

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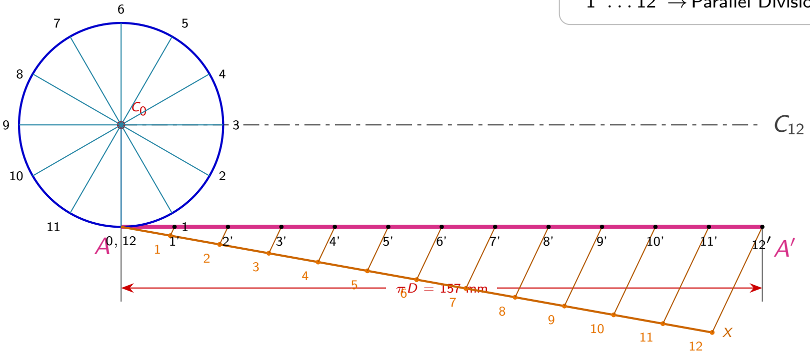

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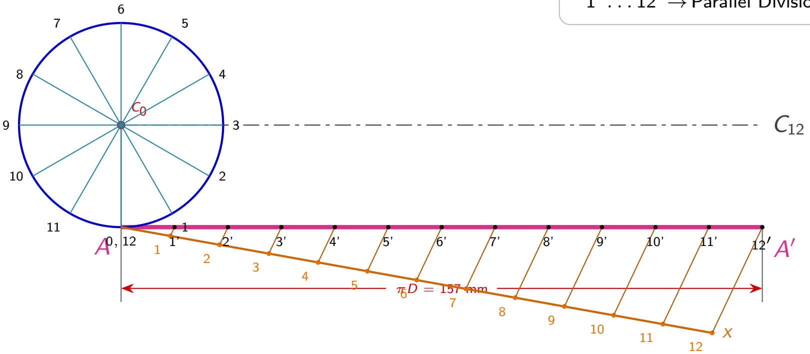

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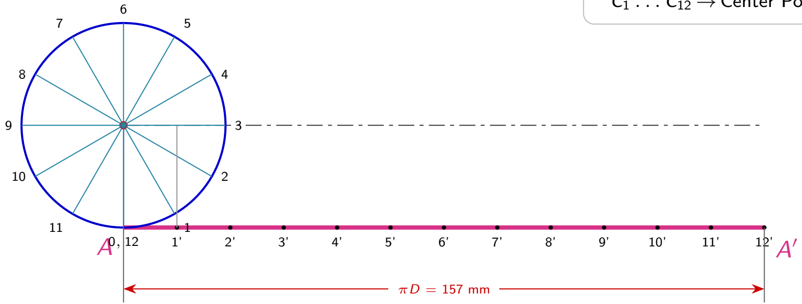

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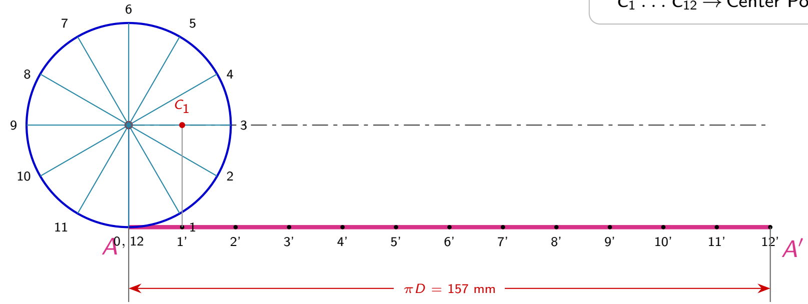

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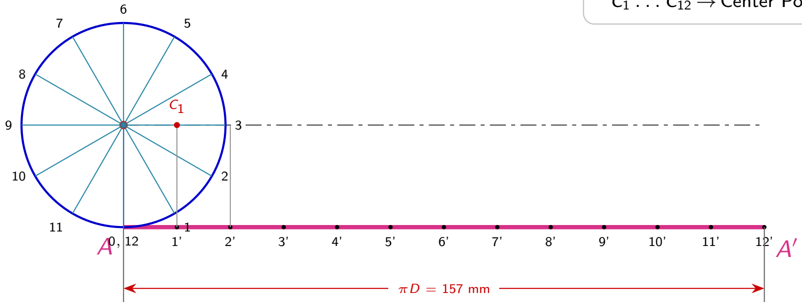

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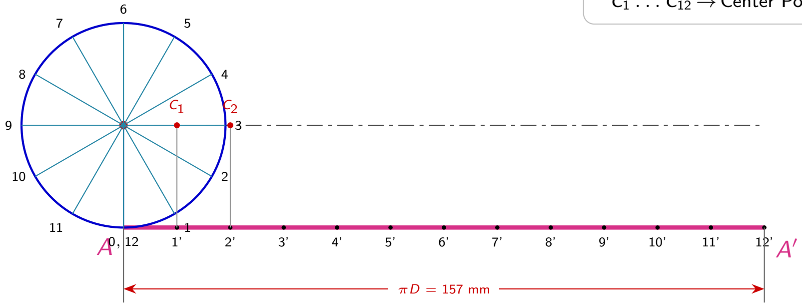

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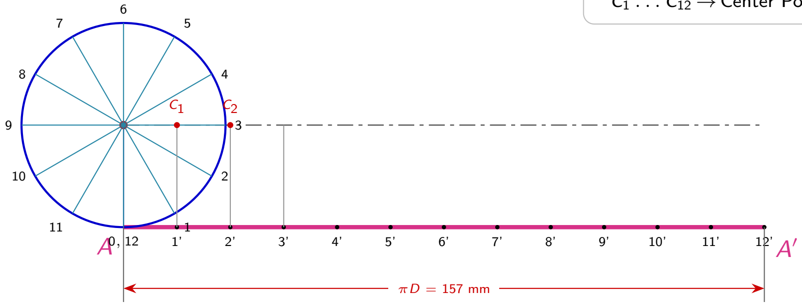

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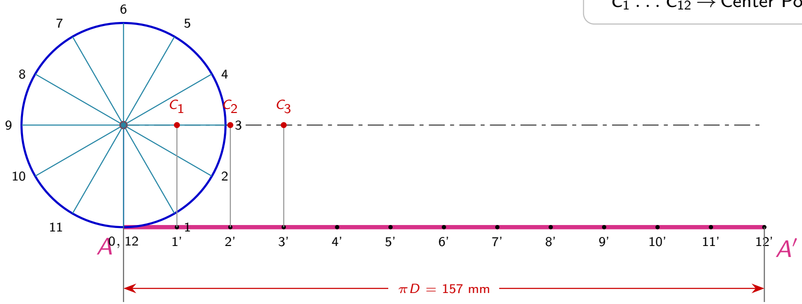

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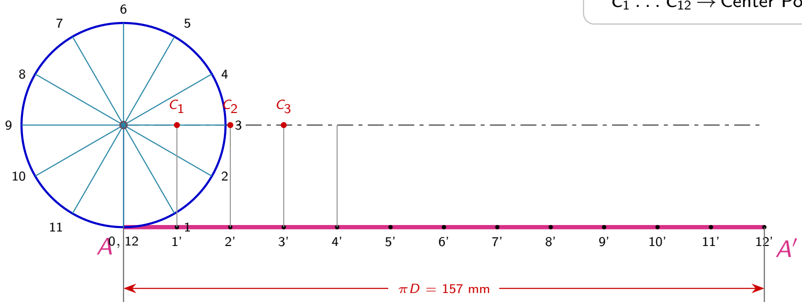

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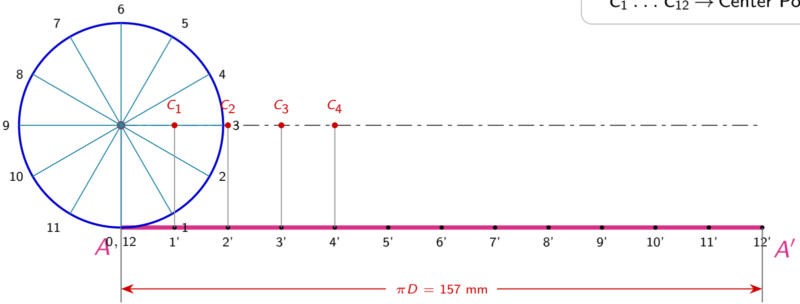

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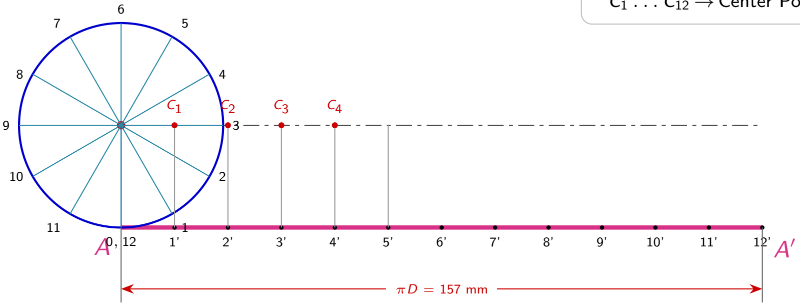

AA'	$\rightarrow$	Directing Line
Axis Line	$\rightarrow$	Center Line
$C_1 \ldots C_{12}$	$\rightarrow$	Center Points

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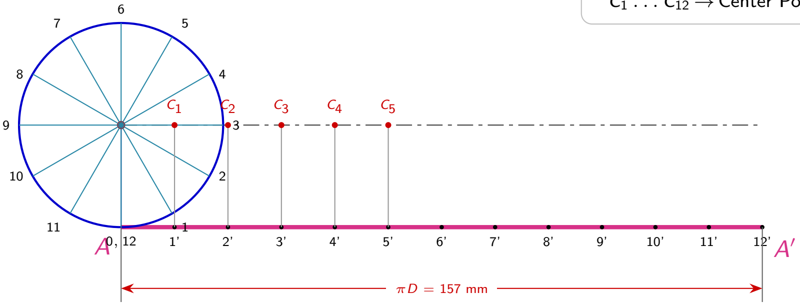

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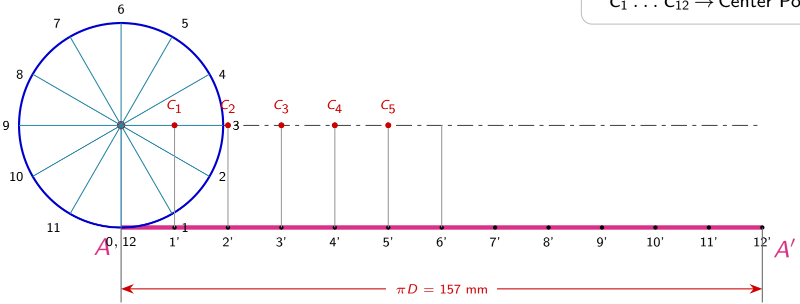

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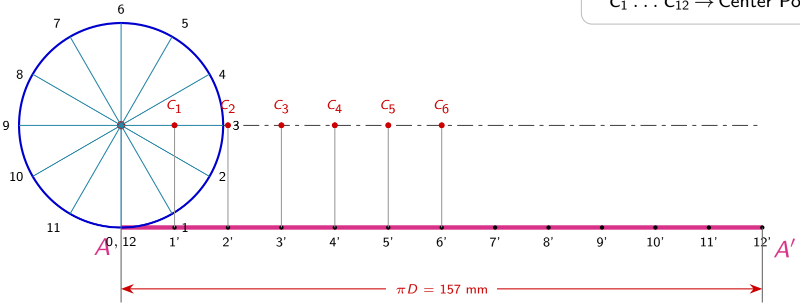

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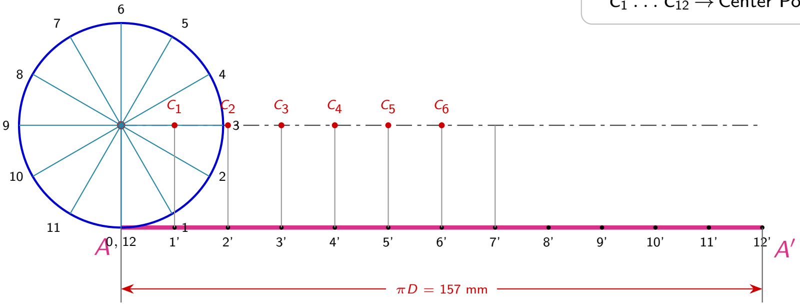

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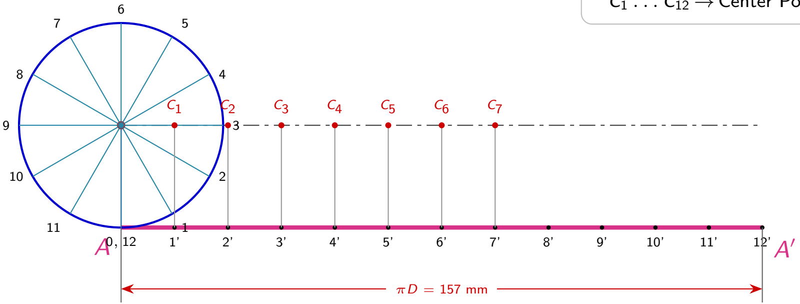

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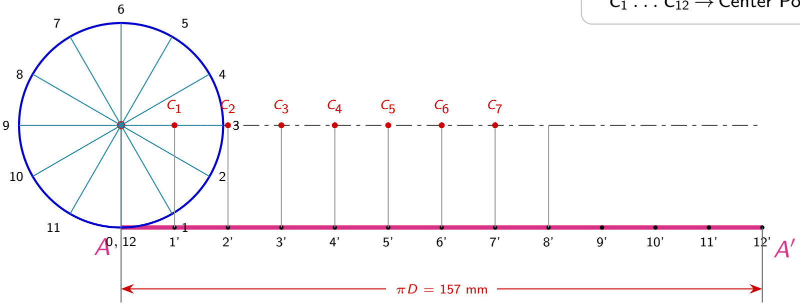

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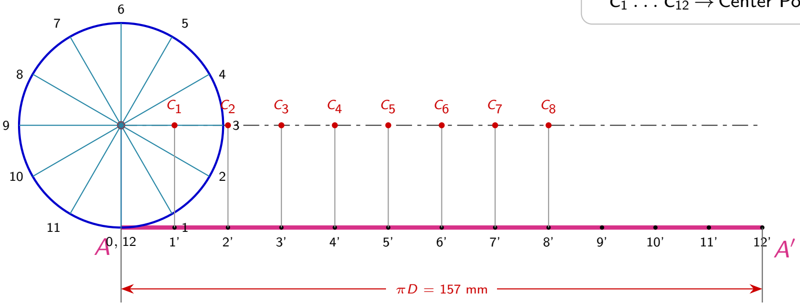

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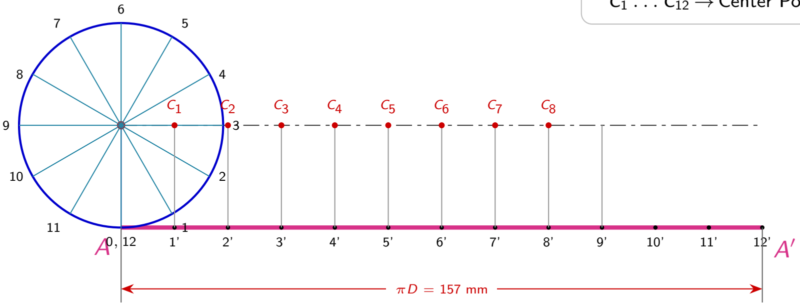

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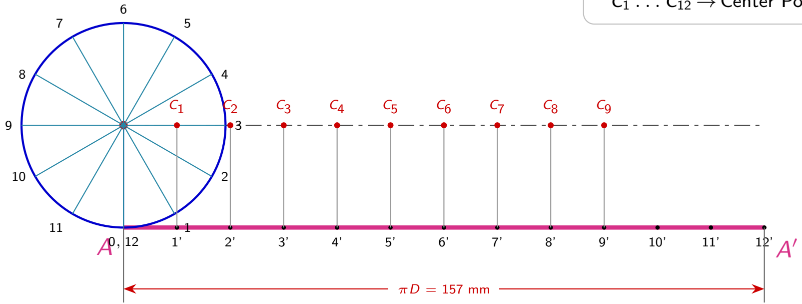

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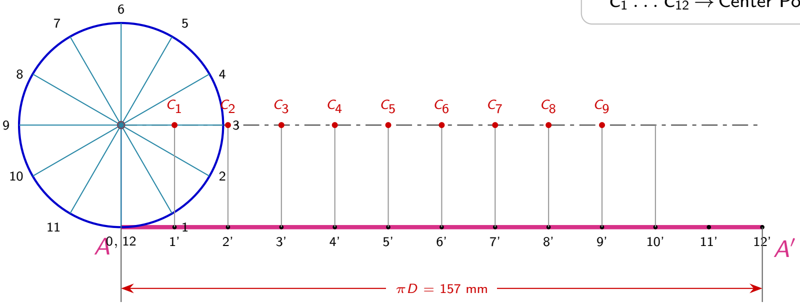

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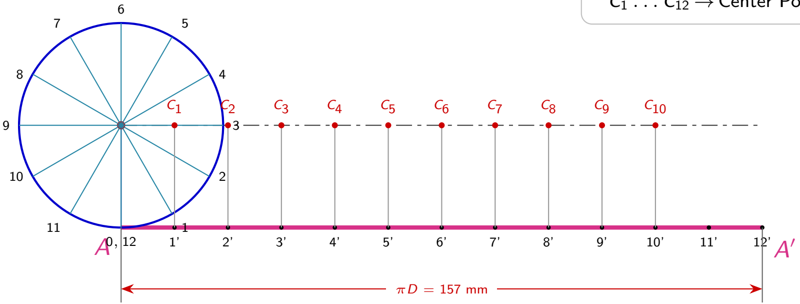

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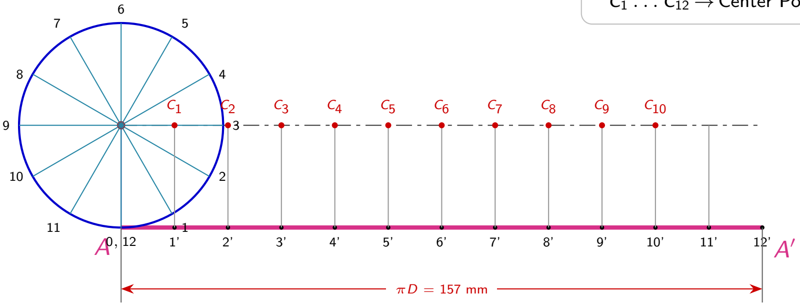

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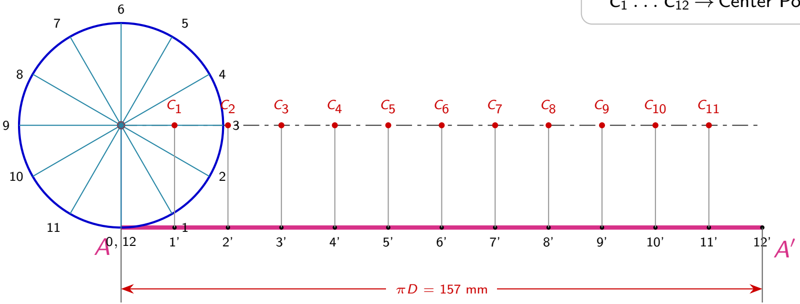

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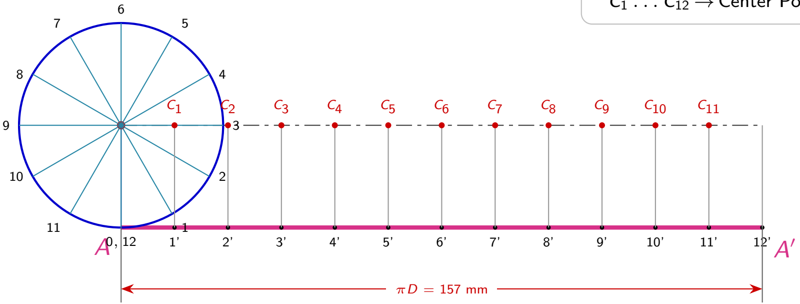

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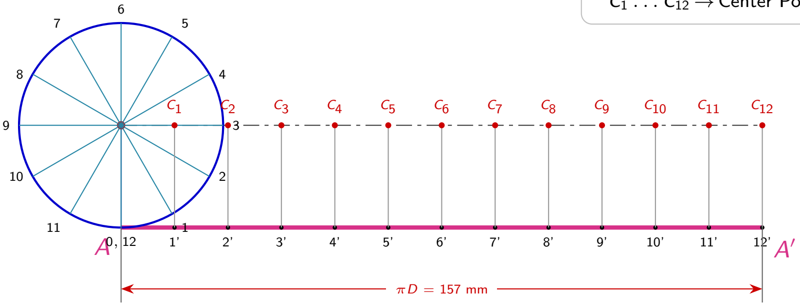

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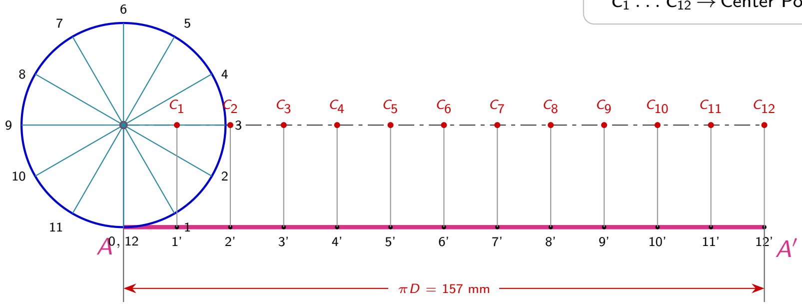
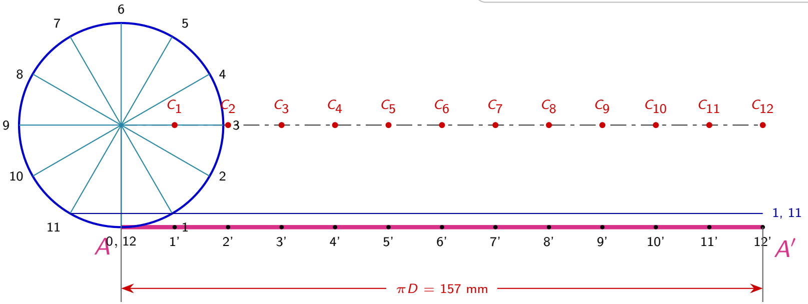


Step 5: Draw Horizontal Reference Lines

Blue Lines $\rightarrow$ Horizontal Reference Lines
$C_1 \ldots C_{12}$ $\rightarrow$ Retained Centers



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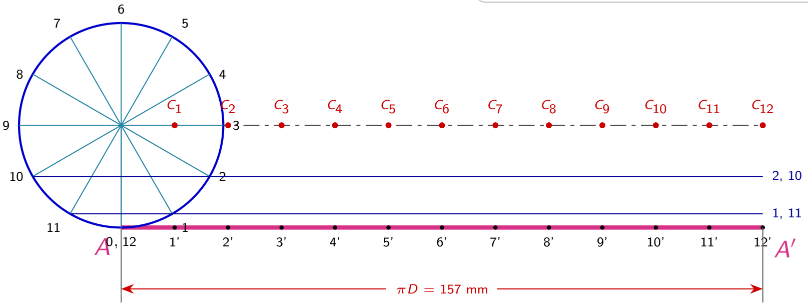

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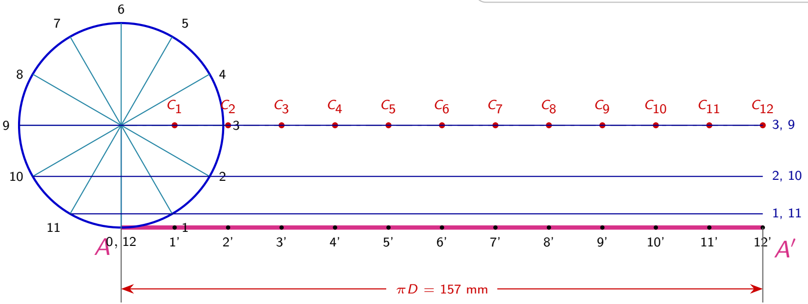

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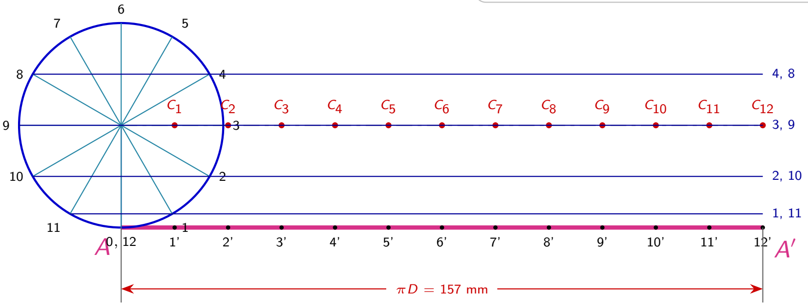

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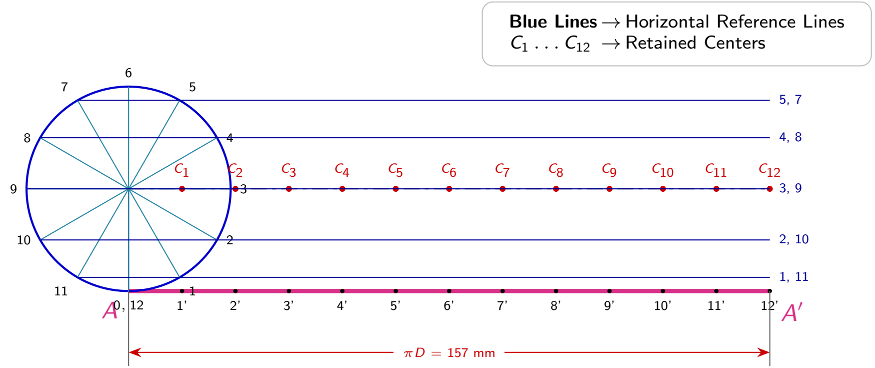

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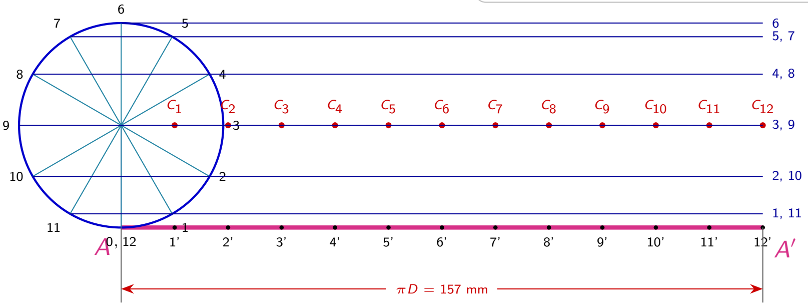
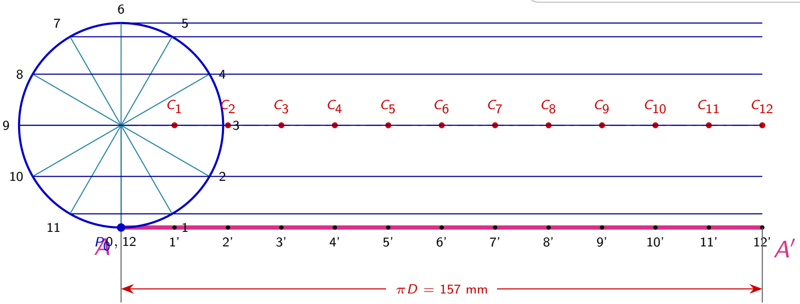


Step 6: Strike Cutting Arcs to Locate Points $P_1 \ldots P_{12}$

$C_1 \ldots C_{12} \rightarrow$ Retained Centers
$P_1 \ldots P_{12} \rightarrow$ Cutting Arcs/Points



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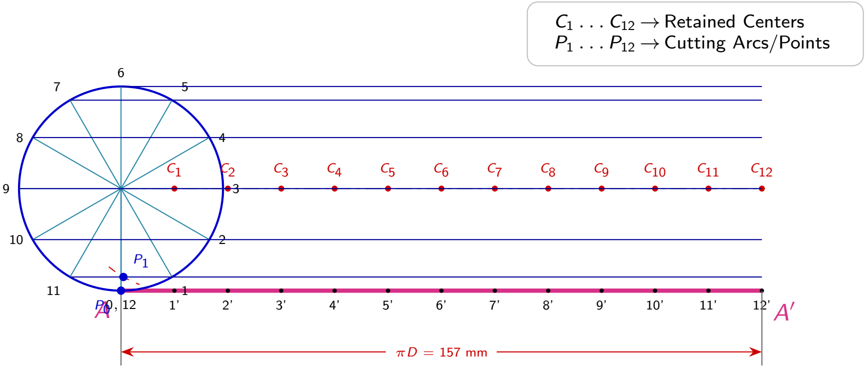

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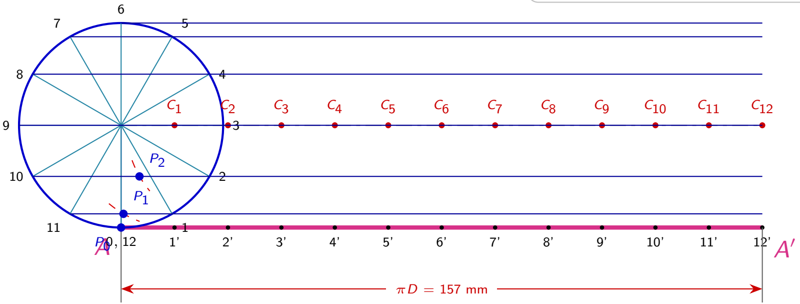

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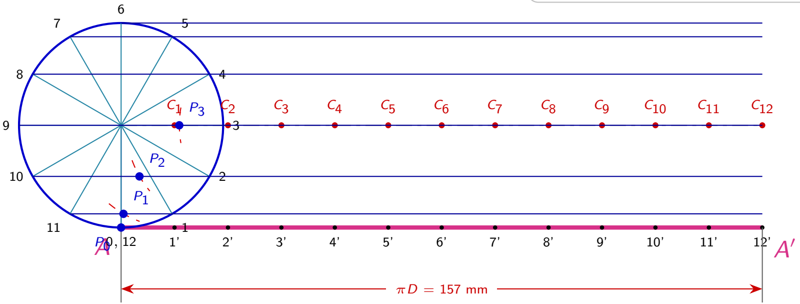

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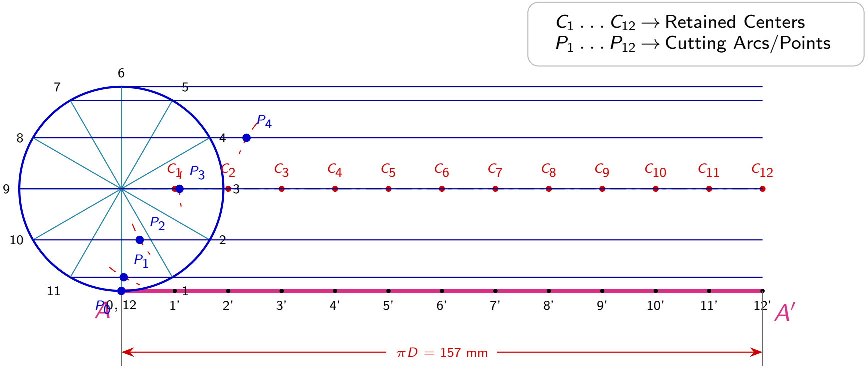

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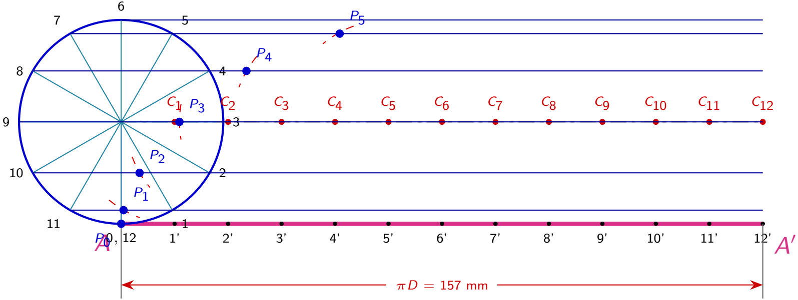

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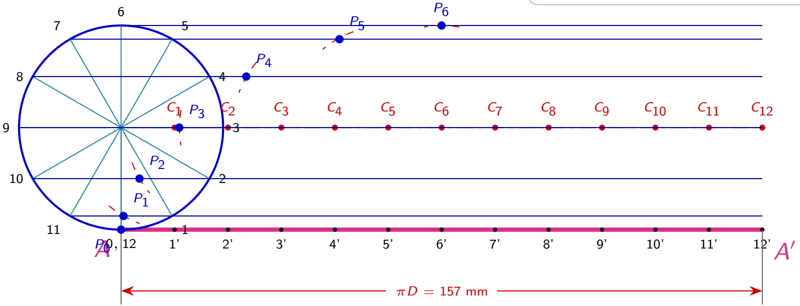

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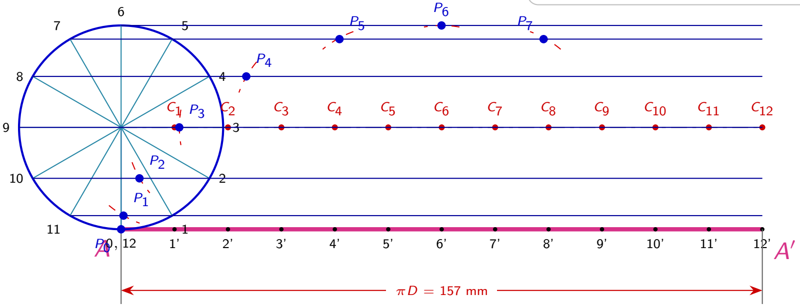

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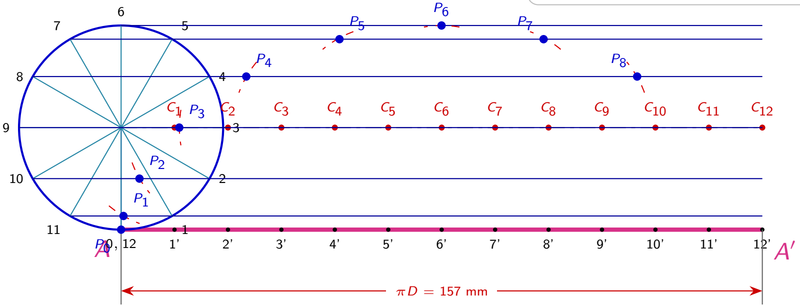

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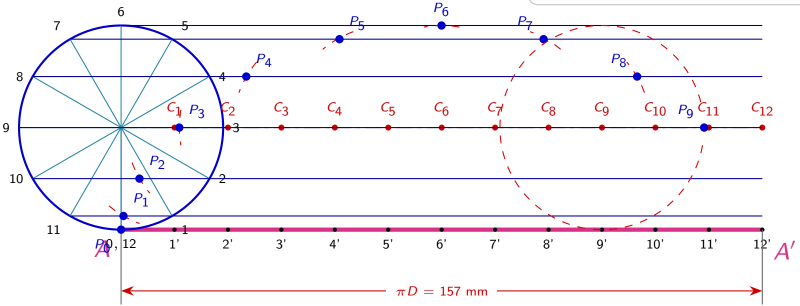

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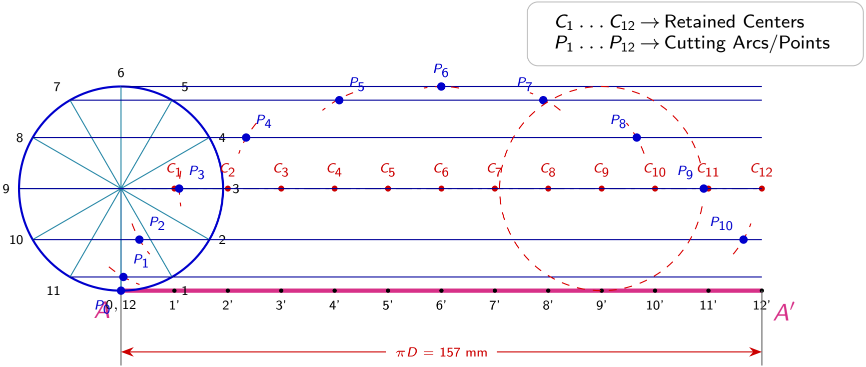

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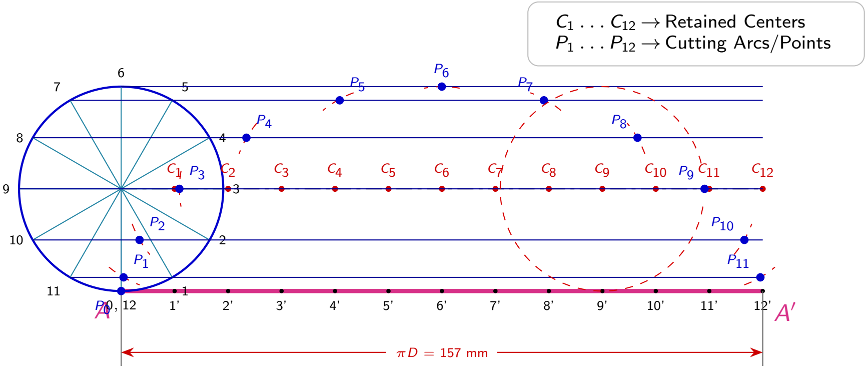

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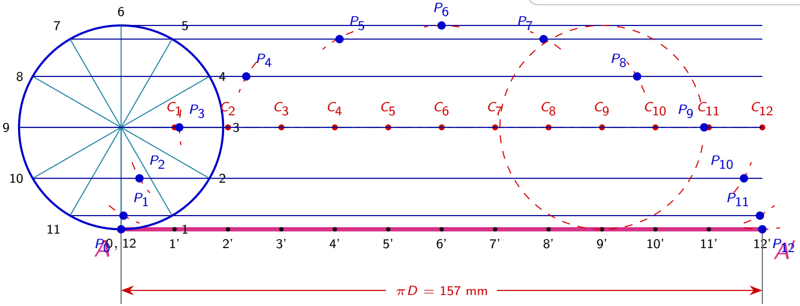
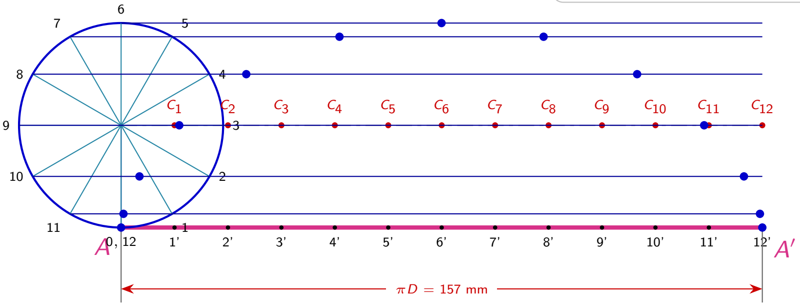


Step 7: Smoothly Join Points to Form Cycloid Curve

Curve $\rightarrow$ Completed Profile
$P_0 \ldots P_{12} \rightarrow$ Retained Points



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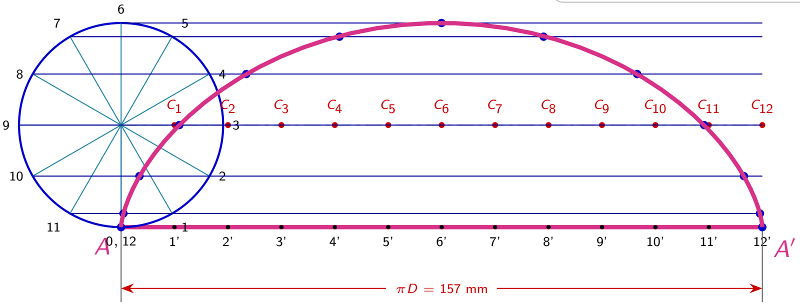
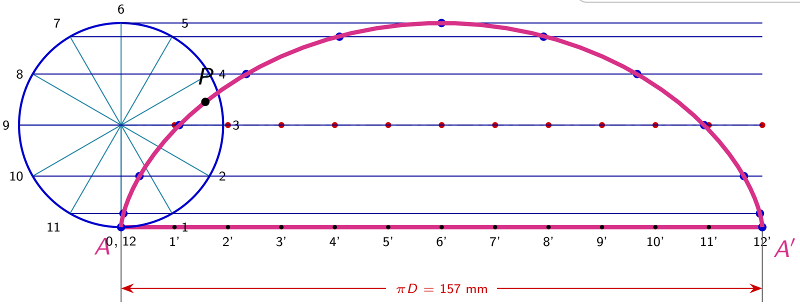


Step 8: Mark Point P & Locate Center C_p

$P \longrightarrow$ Target Point
$C_p \longrightarrow$ Corresponding Center



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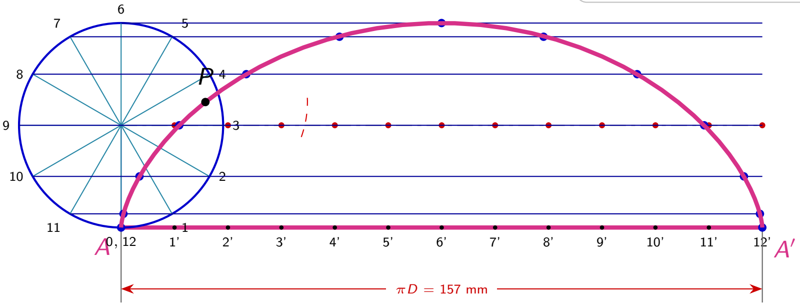

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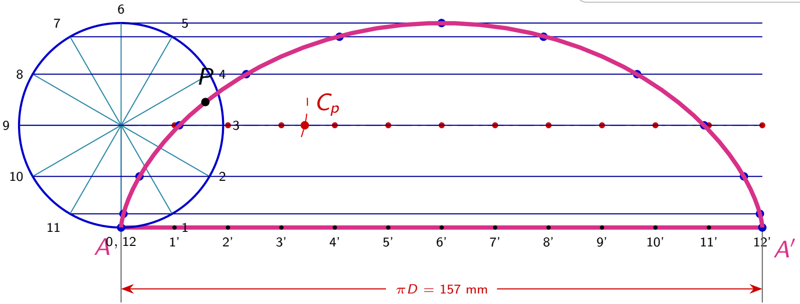

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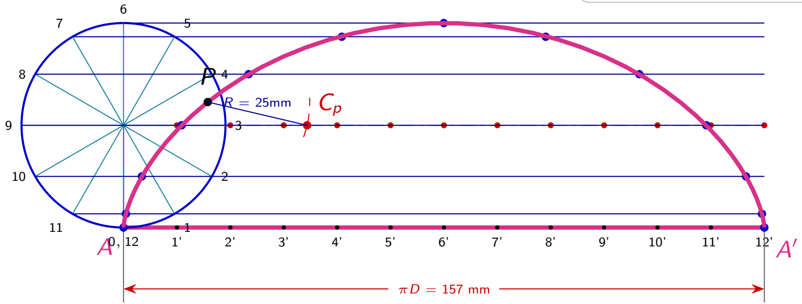
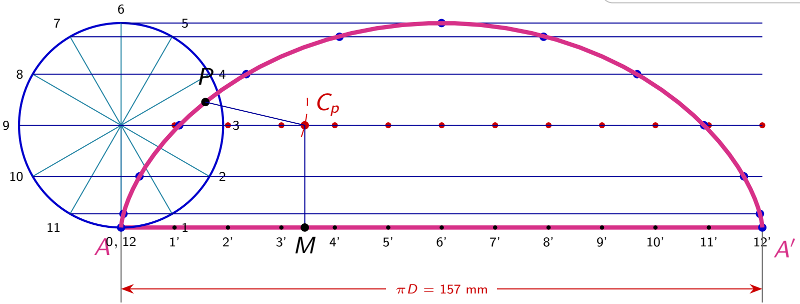


Step 9: Construct Normal Line N-N'

N-N' $\rightarrow$ Normal Line
M $\rightarrow$ Base Projection



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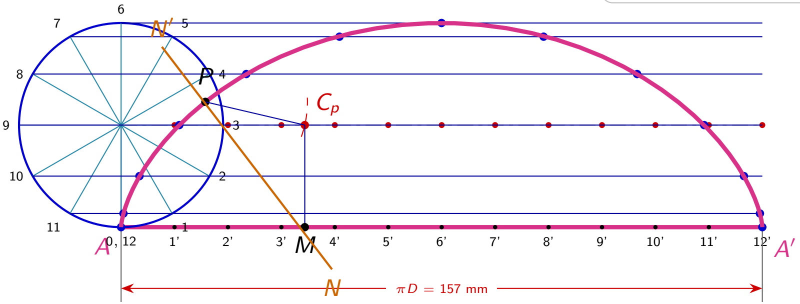
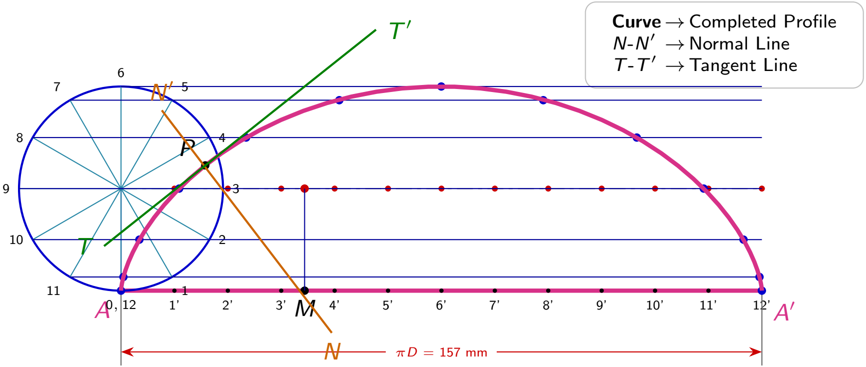


Step 10: Construct Tangent Line T-T' & Final Result



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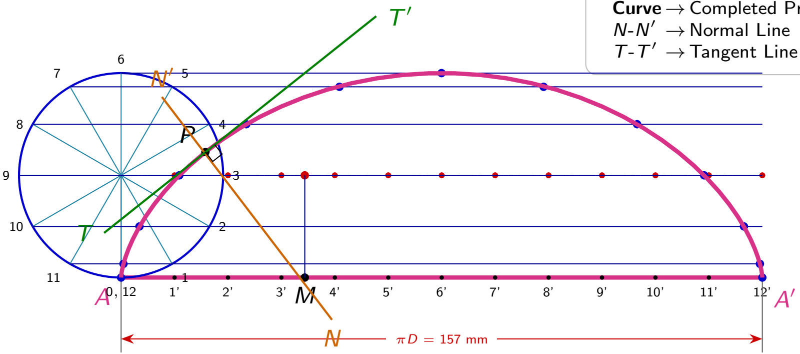

Curve → Completed Profile
N-N' → Normal Line
T-T' → Tangent Line